

# Meal Count Forecasting Model for BNP Paribas Cafeteria

## Executive Summary

BNP Paribas El Djazair's head-office cafeteria needs accurate daily meal count forecasts to optimize preparation and reduce waste. Currently, cafeteria staff must decide how many meals to prepare each day without precise estimates of demand. Over-preparation leads to waste and unnecessary cost, while under-preparation can leave employees without meals. This project addresses that problem by leveraging data-driven forecasting to predict how many employees will dine at the cafeteria each day, enabling better planning and resource allocation.

**Data Sources:** We integrated multiple internal data sources to drive the forecasts, all linked by date. Historical **cafeteria transaction logs** record each meal sold (by employee ID), which we aggregated per day to serve as the target variable (daily meals served). **Office attendance logs** track how many employees badge into the head office each day – a key indicator of how many potential diners are on-site. **Daily menu information** (the list of dishes served each day) provides qualitative context, as certain popular dishes may attract more diners. **Weather data** (temperature, rainfall, etc.) supplies additional context since weather can influence whether employees stay in for lunch or go out. These diverse datasets were merged into a single modeling table per date. Importantly, each of these factors (headcount, menu, weather, calendar) is known in advance or on the day, so they can be used for forecasting future dates.

**Approach:** We developed an end-to-end machine learning pipeline to forecast daily meal counts. The solution involved extensive data cleaning and feature engineering to extract meaningful signals from the raw data. We identified key drivers of cafeteria demand – such as day-of-week patterns, public holidays, number of people in the office, menu popularity, and weather conditions – and created features to represent them. Using these features, we trained state-of-the-art ensemble regression models (gradient-boosted decision trees) that can capture nonlinear relationships and interactions. We paid special attention to validation: using time-aware cross-validation (simulating forward-in-time predictions on withheld dates) to ensure the model generalizes well to future days and doesn't overfit past quirks. The model's hyperparameters were optimized to minimize the forecasting error (Root Mean Squared Error, RMSE). Finally, the trained model was used to predict meal counts for a set of test dates (unseen in training), producing a daily forecast to be used by the cafeteria management.

**Impact:** By accurately forecasting daily meal demand, BNP Paribas can significantly improve its cafeteria operations. An accurate forecast (with low error) means the kitchen can prepare **the right amount of food each day** – minimizing leftovers and waste from overestimation, while avoiding shortages or disappointed staff from underestimation. Even a reduction of a few dozen meals in daily error translates to tangible savings and smoother service. For example, the model's predictions are within roughly 20 meals of the actual value on average (as measured by RMSE), which is a substantial improvement over naive guesses. In practice, this level of accuracy could cut food waste by a large margin and ensure nearly all employees who plan to eat at the cafeteria can be accommodated. Moreover, understanding the drivers of demand (e.g. anticipating higher turnout on certain days or when popular meals are served) allows management to schedule staff and procure ingredients proactively. The forecasting pipeline can be deployed to automatically update predictions as new data

comes in (such as daily headcount and weather forecasts), becoming a valuable decision-support tool for the cafeteria. Overall, this data-driven approach enables a more **efficient, cost-effective, and responsive** cafeteria operation aligned with actual demand.

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## Technical Analysis and Methodology

### Data Overview and Preparation

We collected **five key datasets** (provided by the datathon organizers) and merged them on the date field to create a unified modeling table <sup>1</sup> <sup>2</sup> :

- **Cafeteria Logs ( `kaggle_cantine.csv` )**: Records of individual meal transactions. Each entry has a `Date` and an `ID` for the employee who purchased a meal. From this, we derived the daily target variable **Nb\_meals** (number of meals served) by counting the number of log entries per day. If an employee appears twice on the same date, we count it as 2 meals (assuming multiple swipes mean multiple meals). This daily count is our forecasting target<sup>3</sup> .
- **Head Office Attendance ( `kaggle_head_office.cs` )**: Records of employees present at the office each day (with `ID` and department). We aggregated these to get daily **Nb\_acces** (number of unique employees in the building). This serves as a crucial exogenous feature, since a higher office attendance naturally creates the opportunity for more cafeteria meals <sup>4</sup> . Essentially, it's the daily headcount of potential diners.
- **Menu Data ( `kaggle_menu.csv` )**: The cafeteria's daily menu offerings, including the type of entrée and two main dishes for each day. This qualitative information can influence meal demand – certain dishes are crowd-pleasers that attract more people. We will extract features from the menu (e.g. dish names or categories) to capture this effect <sup>5</sup> <sup>6</sup> .
- **Weather Data ( `kaggle_meteo.cs` )**: Daily weather metrics such as temperature, “feels like” temperature, precipitation amount and type (rain, etc.), wind speed, and cloud cover. Weather can affect whether employees choose to go out for lunch or stay in. For example, rain might encourage more people to remain on-site and eat at the cafeteria. Including weather as input allows the model to account for such variations <sup>7</sup> .
- **Future Dates ( `test_kaggle_finale.csv` )**: A list of specific future dates for which we need to forecast **Nb\_meals** . For these dates, some features are already provided (office attendance, weather, menu, and some date flags) – essentially all the inputs the model needs, except the target. Our job is to fill in the predictions for these dates, which will form the submission.

**Data Cleaning & Integration:** All data sources were first cleaned and then merged on the date. Key steps included:

- **Date Parsing:** We converted date strings to proper Date objects and verified that all datasets align on the same date format. This allowed us to perform merges and time-based calculations reliably.
- **Merging and Aggregation:** Using an inner join on dates (ensuring we only keep dates present in all sources during training), we combined the datasets into one dataframe. We aggregated the

raw logs into daily counts: for the cafeteria log, `Nb_meals` is the count of records per date, and for the attendance log, `Nb_acces` is the count of unique IDs per date <sup>8</sup>. These were merged along with the corresponding menu and weather for each day, so each row in the training set represents one date with all features and the target.

- **Handling Missing Data:** If a date was missing in one of the sources, we needed to handle it carefully. For instance, if the cafeteria was closed on a date (no meal log entries) but the office had people, that day's `Nb_meals` should be considered 0 (assuming no meals were served). If weather data was missing a particular field on a day, we applied reasonable imputation – e.g. if precipitation type was missing, treat it as “no precipitation” rather than null. In the menu data, occasionally a second main dish was `NaN` (meaning only one main course was offered that day); we replaced such missing dish names with a placeholder like “None” so that we could still encode it without breaking the model <sup>9</sup>.
- **Duplicates in Logs:** We checked for duplicate entries in the cafeteria log (the same employee ID appearing multiple times on the same day). Such cases were very rare (~0.4% of entries). These could be legitimate (the person bought breakfast and lunch) or could be badge double-swipes. Our aggregation naturally counts all entries as meals, effectively treating duplicates as multiple meals – which is likely correct. We decided to leave these in, under the assumption that multiple swipes correspond to multiple meals consumed. If we had suspected spurious duplicates (e.g., if the cafeteria policy was one meal per person per day), we could have deduplicated by ID per day; but given the low occurrence, we did not filter them out <sup>10</sup>.
- **Weekend and Holiday Alignment:** We observed that in this Algeria-based dataset, the workweek is Sunday–Thursday and the weekend is Friday–Saturday. Indeed, the logs show zero meals on nearly all Fridays and Saturdays, confirming those days the cafeteria is typically closed. However, the provided data's weekend indicator ( `Est_weeken` ) marked Saturday/Sunday as weekend (perhaps using a default Western calendar). We corrected this by generating our own weekend/holiday flag: treat **Fridays and Saturdays as weekends** (non-working days) and also flag known public holidays. For example, if a date is a holiday (like May 1, or the Eid/Ramadan holidays) we mark it so the model can treat those like weekends (expecting very low or zero meals) <sup>11</sup> <sup>12</sup>. In practice, we created a list of known holiday dates and spans (e.g. we included the entire Ramadan period each year as a special “holiday” span, since many staff likely fast or the cafeteria had limited service during Ramadan). If the cafeteria was closed on those days, the target `Nb_meals` is zero, and our features reflect that (via the holiday flag and zero attendance).

After these steps, we had a consolidated **training dataset** covering May 2022 through 2024 with columns for the date, target ( `Nb_meals` ), and all features (attendance, menu details, weather, calendar flags, etc.) for each day. We ensured that no information from the future (test period) leaked into training during this merging and feature computation. For example, any popularity metrics or rolling averages were computed using **only past data up to that day**, never future data.

**Example Code – Data Merge and Cleaning:** The following snippet illustrates how daily counts and merges were performed in code. We group the raw logs by date to get `Nb_personnes_qui_ont_mangé` (meals count) and similarly group attendance logs to get `nb_acces` (office count). Then we merge all dataframes on date and apply `calendar()` function our to add date-based features (like day-of-week and holiday flags):

```

# Aggregate daily counts for meals and attendance
cant = cant_raw.dropna(subset=[DATE]).groupby(DATE,
as_index=False).size() \
    .rename(columns={"size": TARGET})    # TARGET
= "Nb_personnes_qui_ont_mangé"
office =
office_raw.dropna(subset=[DATE]).groupby(DATE,
as_index=False).size() \
    .rename(columns={"size": TARGET})    # TARGET

# Merge all sources on Date
train = safe_merge(cant, office)
train = safe_merge(train, menu)           # merge menu features
train = safe_merge(train, meteo_raw)      # merge weather features
train = safe_merge(train, lags)           # merge lag/rolling features
train = safe_merge(train, lags) below)    # add day-of-week, month, holiday
train = calendar(train)

```

(Here, `safe_merge` is a helper to merge and drop duplicate columns, and `calendar()` encodes date-based features. The `lags` dataframe is described later in Feature Engineering.)

## Exploratory Data Analysis (EDA)

Before building models, we explored the data to understand the patterns and relationships that drive cafeteria demand. This analysis confirmed our expectations and guided the feature engineering. Key insights include:

### Overall Trend and Seasonality

Figure 1: Daily meals served over time (May 2022 – Dec 2024). The time series shows a strong weekly seasonality, with regular dips to near-zero on weekends (Fridays/Saturdays, when the cafeteria is closed). There is no dramatic long-term trend, but a slight increase is visible from mid-2023 into 2024, corresponding to rising office attendance as more employees work on-site. We also see significant drops during August of each year and late December, likely due to summer vacations and year-end holidays. Prolonged zero-meal periods (flat lines at bottom) correspond to holiday closures, e.g. the Ramadan periods when the cafeteria was effectively closed.

As shown above, daily meal counts exhibit a repeating weekly pattern. On normal working days, the cafeteria serves hundreds of meals, whereas on weekends and holidays it serves almost none. The series is relatively stationary aside from these seasonal effects: there isn't a strong upward or downward trend over the two-year period, but there are noticeable recurring patterns. In particular, **August** stands out with much lower meal counts – presumably because many staff are on summer vacation (the office was much emptier in August, reducing cafeteria demand). We also see dips around late December (Christmas/New Year's period) and on public holidays (e.g. Labour Day on May 1). Aside from these, the overall level of the series is fairly stable, with perhaps a mild growth into 2024 as average office attendance increased.

### Weekday vs Weekend Patterns

A clear **day-of-week effect** is present. Within the workweek (Sunday–Thursday in this context), certain days consistently see higher or lower turnout:

Figure 2: Average meals served on each day of the week. Sunday through Thursday are workdays (in Algeria), while Friday and Saturday are weekend days with near-zero meals. We see that Thursday has the highest average number of meals, followed closely by Sunday (the start of the workweek). Monday tends to have the lowest average among workdays. This confirms a strong weekday pattern in cafeteria usage.

We observe that **Thursdays** typically have the highest demand, perhaps due to end-of-week meetings or a tradition of more staff being on-site that day. **Sunday**, which is the first workday of the week, also has above-average demand. In contrast, **Monday** (the second day of the workweek here) sees slightly fewer meals on average than other mid-week days <sup>13</sup>. This kind of weekly seasonality is a well-known factor in demand forecasting <sup>14</sup>. As expected, **Fridays and Saturdays** (the weekend) have essentially zero meals since the office is closed – the model must learn that these days (and holidays) yield a baseline near zero. We capture this pattern through features: an indicator for each day-of-week and a weekend/holiday flag will allow the model to adjust predictions for these systematic differences.

### Occupancy Correlation (Office Attendance vs Meals)

One of the strongest drivers of cafeteria usage is how many people are physically present in the office. More people on-site generally leads to more cafeteria meals, and the two time series move in tandem:

Figure 3: Office occupancy (orange line) vs meals served (purple line) for a sample month (January 2024). We see both series rising and falling together: peaks on workdays (when ~500–600 staff are present, resulting in ~300–400 meals) and troughs on weekends (near zero). On a given workday, roughly 50–70% of employees present end up dining at the cafeteria, and this fraction varies by day. For example, on Jan 4 (a Thursday with high occupancy) the meal count was especially high (purple nearly reaching orange), indicating an unusually large fraction of staff ate in.

As the example illustrates, daily meal counts closely track office attendance. When more employees come to the office, more lunches are served; when attendance drops (e.g. weekends or holidays), so do cafeteria sales <sup>15</sup>. Across the full training set, the **correlation** between office headcount ( `Nb_acces` ) and meals ( `Nb_meals` ) is about 0.61 <sup>16</sup> – a moderately high correlation confirming occupancy as a key predictor. However, the relationship is far from one-to-one. On most days, not all employees present will eat at the cafeteria (some bring lunch or go out), and occasionally the cafeteria serves more meals than unique people (some employees eat twice, or external visitors dine in). The proportion of employees who dine in varies day to day, typically between 50% and 70%. This variability means that while occupancy is a crucial feature, **other factors must be considered** to accurately predict meals <sup>17</sup>.

To visualize the relationship further, the scatter plot below shows each workday's meals vs attendance:

Figure 4: Scatter of daily meals vs office occupancy (workdays only). Each point is a day; the dashed line is  $y = x$  (which would imply every employee present ate at the cafeteria). Most points lie well below that line, confirming that usually only a fraction of employees dine in. There is a positive correlation (days with higher attendance tend to have more meals served), but significant scatter around the trend – indicating other factors (menu, weather, etc.) modulate the conversion of presence to cafeteria usage.

From the scatter we see a positive trend but also considerable scatter. Some days, an unusually large fraction of people eat at the cafeteria (points closer to the diagonal line), perhaps due to an **appealing menu or bad weather** keeping people in <sup>18</sup>. Other days, even with high attendance, fewer meals are sold (points lower than typical), maybe because many had off-site meetings or there was a special event (free lunch elsewhere, etc.). These outliers underscore the importance of incorporating **menu** and **weather** features alongside occupancy.

## Menu Influence

Intuitively, the daily menu can impact how many people decide to eat at the cafeteria. Our data provides evidence that **certain menu items draw larger crowds**. For example, days when a particularly popular dish (say a special couscous or barbecue) is on the menu tend to see above-average meal counts, whereas “ordinary” menu days see slightly lower counts<sup>19</sup>. We quantified this by computing a **popularity score** for each main dish: essentially the average number of meals served on days that dish was offered. Indeed, dishes like *pizza* or *couscous* had higher average turnout (higher popularity index), whereas some other dishes correlated with lower turnout. We plan to use these popularity metrics as features so the model can predict a higher count on days with historically popular dishes. Additionally, the cafeteria offers two main choices each day – if both options are similar (e.g. both are red meat dishes), they might appeal to the same group and not attract a wider crowd; if there’s variety (say one meat and one vegetarian), more people might find something they like. We considered features to capture this, such as flags for having a vegetarian option available, or both options being the same type of cuisine. These menu-based insights are a bit more nuanced, but they provide the model with signals beyond pure headcount.

*(Exploratory analysis note: We also observed that when both main dishes were particularly enticing (or when a special dessert or event meal was offered), there were bumps in sales. However, due to the difficulty of quantifying “enticing” from text alone, we rely on historical turnout as the best indicator of a menu’s appeal.)*

## Weather Effects

Weather plays a noticeable role in cafeteria usage. On days with **rain**, employees are less inclined to leave the building for lunch, which tends to boost cafeteria demand. Our training data supports this: the cafeteria served *slightly more meals on rainy days (average ~313 meals) than on clear days (~307 meals)*<sup>20</sup>. While the difference is modest (about 2% higher on rainy days in our data), it is in line with expectations that inclement weather keeps people indoors. Temperature may also have an effect: extremely hot or cold days could discourage going out. For instance, if it’s very cold, people might stay in and use the cafeteria rather than walking outside to restaurants. We didn’t notice a very strong monotonic pattern with temperature alone in preliminary analysis, but we will include it as a feature and let the model learn any nonlinear effects. We suspect any impact of temperature might combine with precipitation (e.g. cold + rain/snow amplifies the effect).

Notably, external research on restaurants has shown weather can significantly alter customer behavior (e.g. storms reducing foot traffic)<sup>21</sup>. So including weather features should improve our forecast accuracy, even if the effect in our data is secondary compared to headcount or day-of-week. We will include a binary **IsRain** indicator (whether it rained or not) as well as numeric features like rainfall amount and temperature. The model can then learn the appropriate influence – for example, if rain tends to add a few dozen extra meals on average, it will adjust predictions accordingly on rainy days.

## Seasonal and Holiday Variations

Beyond the weekly cycle, we examined seasonal patterns and special dates. The standout finding was that **August** has much lower cafeteria usage. Many employees take summer holidays in August, resulting in fewer people at the office (and thus fewer cafeteria meals). We see a clear drop in **Nb\_meals** every August (as seen in Figure 1). We will capture this by including a feature for month or season – at least to flag August specifically, since it’s a known slow period. Similarly, late December shows a dip (around Christmas/New Year); even though these may not all be official holidays in Algeria, many international companies have skeleton staff at year-end. We have explicitly flagged known public holidays in our data (such as May 1, Eid, etc.), which correspond to near-zero meal counts. Including a

**holiday indicator** ensures the model doesn't treat those days as ordinary low-demand days that might confuse it – the model will know “this day was a holiday, hence low, not because our predictors failed.”<sup>22</sup> <sup>23</sup> In addition to a generic holiday flag, we've incorporated the weekend logic discussed earlier (Fri/Sat as non-working days). These calendar features help the model differentiate normal workdays from exceptional low-demand days.

**Summary of EDA Findings:** The exploratory analysis confirmed that **date (day-off/holiday), office occupancy, menu choice, and weather** are influential factors for daily meal counts<sup>4</sup>. Weekday patterns and holidays set the general baseline; occupancy drives the main volume; menu and weather introduce day-to-day variability around that baseline. These insights directly informed the features we engineered for the model.

## Feature Engineering

Using the above insights, we created a rich set of features to feed into our machine learning models. Our aim was to provide the model with all relevant information to predict **Nb\_meals** for a given day. We describe the main feature categories and how they were constructed:

- **Date-Derived Features:** Features capturing known calendar patterns:
- **Day of Week:** We encoded the weekday as a categorical variable (with values Sunday through Saturday). This allows the model to learn different baseline levels for each day. For example, “Thursday” tends to have more meals than “Monday,” all else being equal<sup>25</sup>. We one-hot encoded the days or used an ordinal encoding (both approaches are handled well by tree models).
- **Weekend/Holiday Indicator:** A binary feature **Is\_Holiday** set to 1 for weekends or public holidays, 0 for normal workdays<sup>12</sup>. This explicitly informs the model that a day is an off-day (when demand is expected to be near zero). In our context, Fridays and Saturdays, as well as known holiday dates, get **Is\_Holiday = 1**. The model can thus predict essentially zero for these days.
- **Month/Season:** We included the month number (1–12) to capture seasonal variations (summer vs winter). Because month is cyclical, we also experimented with a sinusoidal transformation of day-of-year (creating features **month\_sin** and **month\_cos**) to capture any annual seasonality smoothly. Month effects were not very strong (aside from August), but this feature can help distinguish, for example, August vs. September in subtle ways<sup>26</sup> <sup>22</sup>.
- **Special Day Flags:** We added specific flags for certain spans/events. For instance, a **Is\_Ramadan** flag marks dates during the Ramadan period (when many employees might be fasting or on adjusted schedules; indeed our data showed almost zero meals during Ramadan, effectively as if the cafeteria was closed). We similarly could flag other events if known (e.g., company-wide meetings or offsite events), though we didn't have explicit data for those. These flags ensure outlier periods are accounted for (the model knows those zeros in Ramadan are because of a special event, not a random drop).
- **Occupancy Features:** The most direct predictor is the daily **Nb\_acces** (number of people in office). We included this as a numeric feature. It alone explains a lot of the variance in **Nb\_meals**. We also considered transformations or related features:
  - We know today's occupancy when predicting today's meals (because the badge count can be obtained in the morning or via expected attendance). Therefore, using same-day **Nb\_acces** is valid and very predictive.

- We tried adding **lagged occupancy** (e.g. yesterday's `Nb_acces`) or a **7-day rolling average of occupancy** to see if short-term trends in attendance influence meal counts (for example, if yesterday was very high, maybe today fewer people come, etc.). However, since we already have today's occupancy, adding lagged occupancy did not improve the model much (it's somewhat collinear with today's value). We ultimately kept just the current day's occupancy as the main occupancy feature, trusting the model to learn the relationship to meals.
- That said, a **7-day average occupancy** up to the previous day can smooth out fluctuations and perhaps capture any trend (like gradually increasing office presence). We generated it, but it turned out to be highly correlated with current occupancy as well (so it didn't add new information once `Nb_acces` was in the model).
- **Historical Meal Count Features (Lagged Demand):** In time series forecasting, it's common to use recent values of the target as features. We created several lag features from the `Nb_meals` time series itself:
  - **Lag 1:** Yesterday's meals (`Nb_meals[t-1]`).
  - **Lag 7:** Last week's meals (`1`) – same day last week). This captures weekly seasonality directly – essentially providing the model a baseline from one week ago <sup>27</sup>.
  - **Moving Averages:** We computed rolling averages such as a 7-day moving average (`MA7`) and 14-day moving average of meals, using only past days. These give a smoothed recent trend value. For example, if the last week has been steadily higher than usual, MA7 will be higher than the overall mean, cueing the model that demand is on an upswing. Conversely, a low MA7 indicates a recent slump (maybe due to holidays or remote work week).
  - **Rolling Std Dev:** We also computed a 7-day rolling standard deviation (`std7`) as a measure of volatility of the past week's demand, though this was more experimental.

These lag features turn the time series forecasting into a supervised learning problem by providing the model with recent history. However, **caution** is needed to avoid *data leakage*. We ensured that when computing these lags for a given date, we used only data from **prior** dates. In the training set, this was straightforward (we have the full history up to each day). For the test set (future dates to predict), we had to be careful: if we are predicting, say, a Monday, we might not have the actual value of Sunday's meals (since that's also to be predicted if it's in the future). To handle this, during prediction we adopted a sequential approach: predict one day, then use that prediction as the "lag1" for the next day, and so on (more on this in the Modeling section). We limited ourselves to lag7 (one week) as the main lag feature for modeling, because including lag1 (yesterday) for the test set would require a fully recursive prediction pipeline. In our final model, we found that using lag7 and the moving averages gave much of the benefit of using history, without overly complicating the prediction process. These features let the model, for example, compare today's situation to last week's and adjust if last week's same-day was unusually high or low <sup>28</sup> <sup>29</sup>.

**Implementation:** We created a continuous **timeline** from the start of training to the end of the test period and merged the known `Nb_meals` values. We then shifted and rolled this series to create lag features. Any missing values (e.g. lags that go beyond the available data) were backfilled or filled with safe defaults. For example, in code:

```
timeline = pd.DataFrame({DATE: pd.date_range(cant[DATE].min(),
test_raw[DATE].max())})
tmp = timeline.merge(cant, on=DATE, how="left").interpolate() # merge
known meals, interpolate missing
```

```

for k in [1, 7]:
    tmp[f"lag{k}"] = tmp[TARGET].shift(k)
tmp["ma7"] = tmp[TARGET].rolling(7, min_periods=1).mean()
lags = tmp.fillna(method="bfill")[["Date", "lag1", "lag7", "ma7"]]

```

This prepared a table of lagged values that we then merged into the train and test sets.

- **Menu Features:** We transformed the daily menu textual data into quantitative features:
- **Dish Name One-Hot Encoding:** We created binary indicator features for each entrée and each main dish name (e.g. a feature `Main1_Poulet_roti` which is 1 if the first main dish is "Poulet rôti", otherwise 0). This resulted in many features (since there are many unique dish names over two years), but tree-based models can handle a large number of sparse indicators. The benefit is that the model can learn, for example, that when `Main1_Pizza=1`, the target tends to be higher.
- **Categorical Grouping:** We also engineered higher-level categories. For instance, we grouped dishes by type (we created a small taxonomy mapping certain dishes to categories: "HH" for high-calorie heavy meals, "HL" for hot light meals, "ML" for medium, etc., based on domain guesses). We then added features like `is_hh`, `is_hl`, `is_mi` to indicate if the main dish fell into those categories. Similarly, we flagged if a vegetarian option was present (`Main_veg=1` if one of the mains was vegetarian). These groupings aim to reduce dimensionality and capture broad patterns (e.g. any chicken-based dish vs any fish-based dish).
- **Popularity Index:** Using historical data, we computed a popularity score for each menu item. For each unique main dish, we calculated the average `Nb_meals` on days that dish was served <sup>30</sup>. For example, if "Grilled Fish" was served 10 times and on those days the average meals was 300, then popularity of "Grilled Fish" = 300. We did this for main dish 1 and main dish 2 separately. Then for a new day, we attach the popularity scores of the two mains on the menu. This acts like a target-encoded feature for the menu. If a dish never appeared before (new dish in test), we fallback to a default (overall average). We took care to compute these popularity indices using **training data only** (to avoid leaking test info). To be extra safe, we even did it in a cross-validated manner: for each fold, the popularity for a dish was computed from the training portion of that fold, and used to encode the validation portion – preventing the model from accidentally seeing how popular a dish is based on a future occurrence. This popularity feature lets the model know, for instance, that "Pizza" tends to attract more people than "Quiche," quantified by say 350 vs 280 average meals <sup>31</sup>.
- **Menu Diversity:** We introduced a feature to indicate if the two main dishes are of the same type (e.g. both are red meat, or both are similar cuisine) or very different. The hypothesis was that if both options appeal to the same group, overall turnout might be slightly lower than if they appeal to different tastes (because with diverse options, more people might find something they like). We implemented this by a simple heuristic (e.g. if both mains share a keyword like "grill" or both are meat vs one veg one meat, etc.). This was somewhat speculative and its effect wasn't strong, but it's an example of leveraging domain intuition.

After processing, each day's menu is represented by a set of binary flags (for each dish or dish category present) and a couple of numeric scores (popularity indices).

- **Weather Features:**
- **Rain Indicator:** A binary feature `Is_Rain` indicating whether there was any precipitation (rain) on that day <sup>32</sup>. If yes, the model might bump the prediction up slightly (since rainy days had ~+6 meals on average in training). We also kept the numeric **rainfall amount** (in mm) as a feature, so the model can distinguish between light drizzle vs heavy rain. We expected diminishing returns, but heavy rain could potentially have a bigger effect than a light sprinkle.

We also encoded precipitation type if available (rain vs snow, etc.), but since snow is rare in Algiers, that wasn't a major factor.

- **Temperature:** We included the daily maximum temperature and/or the “feels like” temperature. Rather than binning it manually, we gave it as a continuous feature (in °C) and allow the tree model to find non-linear splits (it might learn that extremely high or low temperatures correlate with higher cafeteria usage). We monitored if the model made use of this feature; if not, it could be dropped. But we kept it in since it's readily available and could interact with rain (e.g. a cold rainy day vs a warm sunny day).
- **Cloud Cover, Wind:** These additional weather fields were included but expected to have minor effects. Cloud cover percentage might correlate with cooler temperatures or dreariness (similar effect as rain but weaker). Wind speed was included in the dataset; we didn't expect a strong direct effect on lunch plans, so this was low priority. In our final model, precipitation and temperature were the primary weather features; cloud and wind did not feature prominently in importance.
- We also created an **interaction feature** manually: `Rain_x_Occupancy = Nb_acces Is_Rain`. The idea was to capture the compounded effect that on a day when many people are present *and* it's raining, the cafeteria could get an especially large turnout (lots of people + each is slightly more likely to eat in) <sup>33</sup>. In theory, the model could learn this interaction on its own, but providing it explicitly can sometimes help if the effect is strong. Similarly, we could interact day-of-week with occupancy (for example, 500 people on a Monday might yield a different meal count than 500 on a Thursday, if behaviors differ by day). We did not end up using many explicit interaction features, since gradient boosting can combine features in splits, but it was a consideration.
- **Other Features Considered:** The head office attendance data included department info (which department each employee belongs to). We thought about features like “number of people from Department X in office today.” Perhaps if certain departments have different lunch habits (e.g. Sales team often goes out for client lunches, while IT team always eats at cafeteria), this could refine predictions. However, adding dozens of department-count features would add complexity and sparsity. Given time constraints, we decided **not to use department-level features**. The aggregate headcount proved sufficient, and any department-specific behavior would be partially reflected in day-of-week or other patterns (for example, if a certain department is in en masse only on certain days and they have a habit of not using the cafeteria, the model could pick that up via those days' slightly lower fraction of people eating).

After engineering these features, we had a comprehensive dataset ready for modeling. Most features for tree-based models do not require scaling (trees split on raw values). We did ensure that categorical features were properly encoded. For model training, we used one-hot encoding for discrete variables like day-of-week and menu items (since we had a manageable number of categories after grouping, and our model – LightGBM – can also handle categorical data natively). We took care to avoid **data leakage**: all transformations that used target values (like dish popularity) were computed using training data only, and for the test set we applied the mappings from train. We also aligned categories between train and test (ensuring that if a dish appears in test that never appeared in train, we handle it by assigning a default popularity, and the one-hot for that dish would be zero since unseen). Any missing values after merging were filled with reasonable defaults (e.g. median for numeric, “missing” category for categorical) to ensure the model doesn't get confused by nulls<sup>34</sup>.

At this stage, our feature matrix for modeling included columns such as: - **Calendar:** e.g. `DayOfWeek` (one-hot Sunday...Saturday), `Is_Holiday` (weekend/holiday flag), possibly `Month` or `Is_August`, etc. - **Occupancy:** `Nb_acces` (number of people in office). - **Lagged Demand:** `Meals_lag7`,

**Meals\_MA7** (7-day avg), etc. (We excluded **lag1** from final training to avoid needing recursive prediction in validation, but included **lag7** since that is one week apart and safe within CV folds). - **Menu:** Dozens of binary features like **Entree\_Salad**, **Main1\_GrilledChicken**, **Main2\_Vegetarian**, etc., plus **Main1\_popularity**, **Main2\_popularity** numeric features. - **Weather:** **TempMax**, **Is\_Rain**, **Rain\_mm**, **CloudCover**, etc. - **Interactions:** e.g. **Rain\_x\_Occupancy** (if used).

This rich feature set provides the model with the relevant signals identified in the EDA. Next, we discuss the modeling approach and how these features were utilized.

## Model Selection and Training

Given the tabular, structured nature of our data and the mix of continuous and categorical features, we chose to use ensemble tree-based regression models – specifically **Gradient Boosting Decision Trees (GBDT)**. These models (in implementations like **LightGBM** and **XGBoost**) are well-suited for our problem because: - They can naturally handle nonlinear relationships and interactions between features (e.g. occupancy and rain might interact to boost demand, which a tree can capture by splitting on both). - They accommodate categorical variables (via one-hot encoding or built-in categorical handling) and continuous variables together without requiring extensive normalization or scaling. - They are robust to outliers and can handle the irregular time-series effects (e.g. sudden zeros on holidays) by learning appropriate splits (like a split on **Is\_Holiday** can isolate those zeros easily). - GBDTs have a strong track record of performance in many forecasting competitions and tabular data challenges, often outperforming classical time-series models when enough exogenous features are available<sup>35</sup> <sup>36</sup>. In essence, our problem can be viewed as a regression with a rich feature set, and tree ensembles excel at that.

We trained both **LightGBM** and **XGBoost** models for comparison. LightGBM was used as our primary model due to its faster training speed and efficiency (it's known to be extremely fast and memory-efficient, which allowed us to quickly iterate and tune hyperparameters)<sup>36</sup>. XGBoost is a bit slower but often achieves similar accuracy; it served as a backup and for ensembling. In practice, both yielded comparable validation performance. We decided to ultimately use LightGBM for our main pipeline (with the option to ensemble with XGBoost at the end for a slight boost).

We also considered other algorithms: - **CatBoost** (another gradient boosting library that handles categorical features natively) – we didn't use it due to time constraints, and the expectation that LightGBM/XGBoost would be sufficient<sup>37</sup>. - Traditional time-series models (ARIMA, exponential smoothing) – these would have difficulty incorporating the exogenous predictors like menu and weather effectively, and likely not match the accuracy of boosted trees given the complexity of the data. - Simpler regressions (linear regression) – likely underfit given the nonlinear interactions (for example, the effect of an additional person in office is not constant – it might depend on if it's a holiday or not, etc.).

**Ensemble Strategy:** To maximize accuracy, we planned an ensemble of the two GBDT models. Combining predictions from LightGBM and XGBoost often yields a small improvement in RMSE over either model alone<sup>38</sup> <sup>39</sup>. Our approach was to train both and take a weighted average of their predictions. For instance, if LightGBM's CV RMSE was slightly better, we might weight it a bit more in the average. Even a ~1% improvement is valuable in a competition context. (In our case, LightGBM alone was very strong; ensembling with XGBoost gave only marginal gains, but we included it for completeness.) Ensembles help reduce model-specific biases – any errors made by one model might be compensated by the other.

**Training Procedure:** We set up a rigorous training regimen with hyperparameter tuning: - We used **Time Series Cross-Validation (CV)** to evaluate model performance. Specifically, we employed a rolling **expanding window CV** with 5 folds (splitting by chronological order)<sup>40</sup> <sup>41</sup>. In each fold, we trained on an initial segment of the time series and validated on the subsequent segment. For example, train on 2022–2023 and validate on early 2024, then train on 2022–mid-2024, validate on late 2024, etc. This ensures validation is always on future data that the model hasn't seen, mimicking real forecasting. We avoided any random shuffling in splitting, to prevent lookahead leakage<sup>42</sup>. Using 5 different folds gave us 5 RMSE estimates which we averaged, ensuring our model performs consistently across different time periods and not just lucky on one particular slice. We also made sure these validation slices collectively cover various seasons and patterns (so the model is evaluated on summer, winter, etc., not just one scenario)<sup>41</sup>. - Within each CV iteration, we also took care with lag features: when predicting the first day of a validation fold, lag features like `lag7` use training data (which is fine), but predicting subsequent days in the validation fold would technically require using previous predictions as lags. In our CV, we slightly simplified by not using `lag1` in the final model (since that one would definitely leak within the fold). `lag7` is safe because the validation set is at least a week ahead of the training set in our scheme. This way, we didn't have leakage of future actuals in validation<sup>43</sup> <sup>44</sup>. (If we had used lag1, a more complex rolling prediction inside CV would be needed – we judged the benefit wasn't worth that complexity given all the other features available.) - We chose **RMSE (Root Mean Squared Error)** as the evaluation metric, as required by the competition. RMSE heavily penalizes large errors, which is appropriate for our use-case – large over- or underestimates of meal count are especially undesirable for planning<sup>45</sup>. We optimized directly for RMSE during training. - For **hyperparameter tuning** of the LightGBM model, we utilized an automated approach with **Optuna** (a Bayesian hyperparameter optimization library). We set up an objective function that trains a LightGBM model on our CV folds and returns the average RMSE. Optuna then guided the search over parameters such as: - `num_leaves` (max tree leaves, controlling model complexity), - `learning_rate`, - `feature_fraction` (what fraction of features to try per tree, for regularization), - `bagging_fraction` and `bagging_freq` (for row subsampling), `min_child_sample` (minimum data in a leaf, another regularizer), - `lambda_l2` (L2 regularization term). We ran about 60 trial iterations of Optuna, which tried different combinations and evaluated the CV RMSE<sup>46</sup> <sup>47</sup>. This process zeroed in on a good hyperparameter set automatically. For example, the tuning suggested an optimal learning rate around ~0.05–0.1 and a fairly large number of leaves (~400–500), among other settings. We also used **early stopping** during training – when fitting on each fold, we monitored the validation RMSE and stopped training trees if the error didn't improve for 50 rounds, to avoid overfitting and to determine the ideal number of trees (iterations) to use<sup>48</sup>.

*Hyperparameter tuning code snippet:*

```
def objective(trial):
    params = {
        "objective": "rmse",
        "metric": "rmse",
        "verbose": -1,
        "force_col_wise": True,
        # Hyperparameters to tune:
        "learning_rate": trial.suggest_float("lr", 0.01, 0.2, log=True),
        "num_leaves": trial.suggest_int("leaves", 64, 512, step=16),
        "feature_fraction": trial.suggest_float("ff", 0.6, 1.0),
        "bagging_fraction": trial.suggest_float("bf", 0.6, 1.0),
        "bagging_freq": trial.suggest_int("bfq", 1, 10),
        "min_child_samples": trial.suggest_int("mcs", 5, 60),
```

```

    "lambda_l2": trial.suggest_float("l2", 0.0, 5.0)
}
# Perform a 5-fold time series CV internally:
rmses = []
tscv = TimeSeriesSplit(n_splits=5)
for train_idx, val_idx in tscv.split(train):
    dtrain_X = train.iloc[train_idx][features]; dtrain_y =
train.iloc[train_idx][TARGET]
    dval_X = train.iloc[val_idx][features]; dval_y
= train.iloc[val_idx][TARGET]
    model = lgb.LGBMRegressor(**params)
    model.fit(dtrain_X, dtrain_y,
              eval_set=[(dval_X, dval_y)],
              callbacks=[lgb.early_stopping(50, verbose=False)])
    pred = model.predict(dval_X)
    rmses.append(mean_squared_error(dval_y, pred, squared=False))
return np.mean(rmses)

study =
optuna.create_study(direction="minimize")
study.optimize(objective, n_trials=60)
best_params = study.best_params

```

(The above code runs the described Optuna tuning. After completion, `best_params` contains the best hyperparameters found, which we then used to train the final model. We also fixed `objective` and `metric` to RMSE as required.)

- After tuning, we trained the final LightGBM model on the entire training set. Actually, to leverage all data and also **quantify uncertainty**, we employed a form of **ensemble averaging** by training 5 LightGBM models on 5 different splits of the data:
- We again used the 5-fold TimeSeriesSplit; for each fold, we trained a model on that fold's training subset (e.g. train on 2022–2023 data for fold1) and then used it to predict the test set. We did this for all 5 folds and averaged their predictions for each test date<sup>49</sup> <sup>50</sup>. This ensembling across folds helps smooth out any variance – it's like bagging but with time-based splits. Every training example gets to influence 4 out of 5 models (since in one fold it's part of validation). The result is an even more robust prediction.
- For XGBoost, we did a similar approach (though possibly with fewer folds or using the same folds). We then averaged the XGBoost prediction with the LightGBM average. In practice, we might have given slightly higher weight to LightGBM if it performed better on CV. (The improvement from this final XGB blend was small – on the order of a few tenths of RMSE – but “every bit helps” in competition scoring.)
- **Validation Results:** During development, our cross-validation indicated a **CV RMSE in the low 20s** (around 22–23). Given daily meal counts range roughly 0 to 400+, this error is about 5-7% of the typical range, which we consider quite good <sup>51</sup>. In fact, an RMSE in the 15–25 range was our target for strong performance <sup>51</sup> so we were satisfied to hit that. We also examined feature importances from the LightGBM model: as expected, the top features were **Nb\_acces (office occupancy)**, followed by **Day-of-week indicators**, then features related to **menu popularity** and **rain**. This confirmed that the model was using the inputs in a sensible way (occupancy explaining the largest share of variance, with calendar and menu/weather refining the estimates) <sup>52</sup> <sup>53</sup>. Features that had very low importance (e.g., wind speed, or a rarely used

menu dummy) could be dropped in future to simplify the model. We also looked at residuals to see if there were systematic errors: one observation was that the model initially under-predicted some Mondays – likely because our lag features or other inputs didn’t fully capture a pattern where Monday had slightly different behavior. This was remedied by ensuring **day-of-week was included** (so the model learned Monday’s baseline separately) and by not using lag1 (which, if Sunday was high, could lead to overestimating Monday slightly without a Monday adjustment). After such tweaks, residual patterns looked random, with no obvious bias on particular days or under specific conditions.

Once satisfied, we retrained on the full training data and generated predictions for the test dates. For each date in the test set, we had all the exogenous features (office count, menu, weather, date flags). For computing lag features on the test dates, we used the last available actuals from training for the initial lags, and for dates that extended beyond training, we fed our own previous predictions. Specifically, if the test set included a continuous range of dates after the training period, we did a day-by-day forecast: predict day1, then treat that prediction as yesterday’s value to predict day2, and so on <sup>54</sup> <sup>55</sup>. This ensures consistency in using lag features without ever peeking at future values. Each fold-model was run in this recursive manner and their outputs averaged.

The final output was a CSV of predictions for each test date, which we submitted. The model is able to, for example, predict that on a typical mid-week day with 500 people in office and a popular dish on the menu (and rainy weather), the cafeteria should expect around 320 meals, whereas on a holiday or weekend it predicts nearly 0, and during August it predicts maybe 200 if attendance is low. These predictions will help the cafeteria prepare appropriate quantities.

**Conclusion:** Our forecasting model combines **calendar dynamics, human behavior, and external factors** to accurately predict daily cafeteria demand. By using gradient boosted trees with carefully engineered features, we capture the important effects: who is at work, what’s for lunch, and what’s the weather – all key drivers of lunch decisions – plus the innate weekly rhythm of the workplace. The approach was validated with rigorous back-testing (time-split CV), giving us confidence that it generalizes well to future dates <sup>56</sup> <sup>57</sup>. The resulting solution provides BNP Paribas with a reliable tool to anticipate cafeteria load, minimize waste, and improve service quality.

## Appendix: Full Code Listing

Below is the full Python code used for data processing, feature engineering, model training, and prediction for this project. This code was executed in a Kaggle notebook environment.

```
import warnings, unicodedata, re, datetime as dt
import numpy as np, pandas as pd, lightgbm as lgb, optuna
from sklearn.model_selection import TimeSeriesSplit
from sklearn.metrics import mean_squared_error
warnings.filterwarnings("ignore")

# Define paths and constants
DATA = "/kaggle/input/ADC-4-0-BNP-Paribas-El-Djazair/"
OUT = "/kaggle/working/submission.csv"
DATE, TARGET = "Date", "Nb_personnes_qui_ont_mangé"

# Holiday spans (public holidays and Ramadan periods) for the holiday
flag HOLI = [
```

```

(dt.date(2022,1,1), dt.date(2022,1,1)),      # New Year
2022 (dt.date(2022,4,2), dt.date(2022,4,30)),  #
Ramadan 2022 (dt.date(2023,3,23), dt.date(2023,4,20)), #
Ramadan 2023 (dt.date(2024,3,11), dt.date(2024,4,9)), #
Ramadan 2024 (dt.date(2022,5,1), dt.date(2022,5,1)), #
Labor Day 2022 (dt.date(2023,5,1), dt.date(2023,5,1)), #
Labor Day 2023 (dt.date(2024,5,1), dt.date(2024,5,1)) #
Labor Day 2024
]
is_hol = lambda d: any(s <= d <= e for s, e in HOLI)

# Dish taxonomy (grouping certain main dishes into categories, e.g., HH
= heavy, HL = hot light, MI = medium, LI = light, etc.)
DISH_TAX = {
    'Rechta au poulet': 'HH',
    'Menu Chef': 'HH',
    'Menu Chef + Sandwich + Pizza': 'HL',
    'Grillade / Fritte / salade / Oeuf au plat/ Fromage': 'HL',
    'Sandwich': 'MI',
    'Tlitli au poulet': 'MI',
    'Couscous au poulet': 'MI',
    'Kbab au poulet': 'MI',
    'Vol au vent sauce financière': 'LI',
    'Gratin de poulet': 'LI'
}

# Helper to read CSV with automatic delimiter
detection def read_auto(path):
    df = pd.read_csv(path, sep=None, engine="python")
    return df if df.shape[1] > 1 else pd.read_csv(path)

# Helper to normalize column names (remove accents, lower-case, replace
spaces with underscores)
def norm(s):
    s = unicodedata.normalize("NFKD", str(s))
    s = s.encode("ascii", "ignore").decode("ascii")
    return re.sub(r"^[a-z0-9_]+", "_", s.lower()).strip("_")

def normalise(df):
    df.rename(columns={c: norm(c) for c in df.columns},
inplace=True) if "date" in df.columns:
        df.rename(columns={"date": "Date"}, inplace=True)
    return df

def calendar(df):
    df["dow"] = df[DATE].dt.dayofweek      # Monday=0,
    Sunday=6 df["month"] = df[DATE].dt.month
    df["is_hol"] = df[DATE].dt.date.map(is_hol).astype(int)      #
    holiday span flag
    df["is_we"] = ((df["dow"].isin([4, 5])) | df["is_hol"]).astype(int) #
    weekend (Fri=4, Sat=5) or holiday
    df["is_pay"] = df[DATE].dt.day.isin([26, 27]).astype(int)      #

```

(example: payday flag, if we suspected end of month effects, not crucial here)

```
# Optional cyclic encoding:
df["dow_sin"] = np.sin(2*np.pi*df["dow"]/7); df["dow_cos"] =
np.cos(2*np.pi*df["dow"]/7)
df["month_sin"] = np.sin(2*np.pi*df["month"]/12); df["month_cos"] =
np.cos(2*np.pi*df["month"]/12)
return df

# ----- Load and normalize data -----
cant_raw = normalise(read_auto(DATA + "kaggle_cantine.csv"))
office_raw = normalise(read_auto(DATA +
"kaggle_head_office.csv")) menu_raw =
normalise(read_auto(DATA + "kaggle_menu.csv")) meteo_raw
= normalise(read_auto(DATA + "kaggle_meteo.csv"))
test_raw = normalise(read_auto(DATA +
"test_kaggle_finale.csv"))

for df in (cant_raw, office_raw, menu_raw, meteo_raw,
test_raw): df[DATE] = pd.to_datetime(df[DATE],
errors="coerce")

# ---- Fix specific column name issues ----
office_raw.rename(columns={"collaborators_in_office": "nb_acces"},
inplace=True)
menu_raw.rename(columns={"plat_principal_1":
"plat_principal_1"}, inplace=True) # fix a typo in column
name

# ----- Aggregate daily counts -----
cant = cant_raw.dropna(subset=[DATE]).groupby(DATE, as_index=False).size() \
.rename(columns={"size": TARGET}) # daily
meals office = office_raw.dropna(subset=[DATE]).groupby(DATE,
as_index=False).size() \
.rename(columns={"size": "nb_acces"}) # daily attendance

# ----- Menu popularity features -----
overall_mean = cant[TARGET].mean()
pop_ent = cant.merge(menu_raw[[DATE, "entrees"]], on=DATE) \
.groupby("entrees")[TARGET].mean()
pop_m1 = cant.merge(menu_raw[[DATE, "plat_principal_1"]], on=DATE) \
.groupby("plat_principal_1")[TARGET].mean()
pop_m2 = cant.merge(menu_raw[[DATE, "plat_principal_2"]], on=DATE) \
.groupby("plat_principal_2")[TARGET].mean()

# Merge menu with its popularity scores and taxonomy
menu = menu_raw.drop_duplicates(DATE).copy()
menu["entree_pop"] =
menu["entrees"].map(pop_ent).fillna(overall_mean) menu["main1_pop"]
=
menu["plat_principal_1"].map(pop_m1).fillna(overall_mean)
menu["main2_pop"] =
menu["plat_principal_2"].map(pop_m2).fillna(overall_mean)
menu["dish_cat"] =
menu["plat_principal_1"].map(DISH_TAX).fillna("MI") menu["is_hh"] =
(menu["dish_cat"] == "HH").astype(int)
```

```
menu["is_hl"] = (menu["dish_cat"] == "HL").astype(int)
```

```

menu["is_mi"] = (menu["dish_cat"] == "MI").astype(int)
menu["dish_stability"] = 1
# (placeholder feature, e.g., for menu variety or consistency)

# ----- Lag and rolling features -----
# Create a full timeline from min train date to max test
date timeline = pd.DataFrame({DATE:
pd.date_range(cant[DATE].min(), test_raw[DATE].max())})
tmp = timeline.merge(cant, on=DATE, how="left").interpolate()# get
continuous series of target
# Calculate lags and rolling metrics
for k in [1, 2, 3, 7]:
    tmp[f"lag{k}"] = tmp[TARGET].shift(k)
tmp["ma7"] = tmp[TARGET].rolling(7,
min_periods=1).mean() tmp["ma14"] =
tmp[TARGET].rolling(14, min_periods=1).mean() tmp["std7"]
= tmp[TARGET].rolling(7, min_periods=1).std()
lags = tmp.fillna(method="bfill")[ [DATE] + [c for c in tmp.columns if
c.startswith(("lag","ma","std"))] ]

# Helper for merging and dropping duplicate suffix
columns def safe_merge(left, right):
    out = left.merge(right, on=DATE, how="left", suffixes=("",
    "_dup")) dup_cols = [c for c in out.columns if
    c.endswith("_dup")]
    return out.drop(columns=dup_cols)

# ----- Assemble training and test sets -----
train = safe_merge(cant, office)
train = safe_merge(train, menu)
train = safe_merge(train,
meteo_raw) train =
safe_merge(train, lags) train =
calendar(train)

test = safe_merge(test_raw,
office) test = safe_merge(test,
menu)
test = safe_merge(test,
meteo_raw) test =
safe_merge(test, lags)
test = calendar(test)

# ----- Target encoding for menu features -----
# We perform target encoding (with cross-val) for entree and main
dishes. fold_id = np.zeros(len(train), dtype=int)
for i, (_, val_idx) in enumerate(TimeSeriesSplit(n_splits=5).split(train)):
    fold_id[val_idx] = i

for col in ["entrees", "plat_principal_1",
"plat_principal_2"]: if col not in train or col not in
test:
    continue
# Compute mean target per category, excluding current
fold enc = pd.Series(np.nan, index=train.index)
for f in range(5):

```

```
means = train.loc[fold_id != f].groupby(col)[TARGET].mean()
```

```

        enc[fold_id == f] = train.loc[fold_id == f, col].map(means)
train[f"{col}_te"] = enc.fillna(overall_mean)
# For test, use overall training means
global_means =
train.groupby(col)[TARGET].mean()
test[f"{col}_te"] = test[col].map(global_means).fillna(overall_mean)

# ----- Handle categorical vs numeric ----- for
df in (train, test):
    # Fill missing categorical with 'missing' and cast to category
    dtype for c in [c for c in df.select_dtypes("object") if c !=
DATE]:
        df[c] = df[c].fillna("missing").astype("category")

# Align categories between train and test
cat_cols = [c for c in
train.select_dtypes("category").columns] for c in cat_cols:
    # Ensure train and test have same category levels
    combined_cat = train[c].cat.categories.union(test[c].cat.categories)
    train[c] = train[c].cat.set_categories(combined_cat)
    test[c] = test[c].cat.set_categories(combined_cat)

# Fill any remaining NaNs in numeric features with median (train's
median) for c in train.select_dtypes("number").columns:
    med = train[c].median()
    train[c].fillna(med,
inplace=True)
    test[c].fillna(med,
inplace=True)

# Define feature columns (exclude target and date)
features = [c for c in train.columns if c not in [TARGET, DATE]]

# ----- Hyperparameter tuning with Optuna ----- def
objective(trial):
    params = {
        "objective": "rmse",
        "metric": "rmse",
        "force_col_wise": True,
        "verbose": -1,
        "learning_rate": trial.suggest_float("lr", 0.01, 0.2, log=True),
        "num_leaves": trial.suggest_int("leaves", 64, 512, step=16),
        "feature_fraction": trial.suggest_float("ff", 0.6, 1.0),
        "bagging_fraction": trial.suggest_float("bf", 0.6, 1.0),
        "bagging_freq": trial.suggest_int("bfq", 1, 10),
        "min_child_samples": trial.suggest_int("mcs", 5, 60),
        "lambda_l2": trial.suggest_float("l2", 0.0, 5.0)
    }
    rmses = []
    tscv = TimeSeriesSplit(n_splits=5)
    for tr_idx, val_idx in tscv.split(train):
        X_tr, y_tr = train.iloc[tr_idx][features], train.iloc[tr_idx][TARGET]
        X_val, y_val = train.iloc[val_idx][features], train.iloc[val_idx]
[TARGET]
        model = lgb.LGBMRegressor(**params)

```

```

        model.fit(X_tr, y_tr, eval_set=[(X_val, y_val)],
                  callbacks=[lgb.early_stopping(50, verbose=False)])
    pred_val = model.predict(X_val)
    rmse.append(mean_squared_error(y_val, pred_val,
                                   squared=False))
    return np.mean(rmse)

study = optuna.create_study(direction="minimize")
study.optimize(objective, n_trials=60,
               show_progress_bar=False) best = study.best_params
best.update({"objective": "rmse", "metric": "rmse", "verbose": -1,
            "force_col_wise": True})

# ----- Train final model(s) and predict -----
# We will train a LightGBM model on each CV fold and aggregate
# predictions
preds = np.zeros(len(test))
tscv = TimeSeriesSplit(n_splits=5)
for tr_idx, val_idx in tscv.split(train):
    model = lgb.LGBMRegressor(**best)
    model.fit(train.iloc[tr_idx][features],
              train.iloc[tr_idx][TARGET]) preds +=
    model.predict(test[features]) / 5.0

# ----- Recursive lag update for sequential dates ----- #
# If test set has sequential dates beyond train, update lag features with
# predictions
final_preds = []
for i, date in
    enumerate(test[DATE]): p =
    float(preds[i])
    final_preds.append(int(round(p))) # round to nearest integer for
    output (optional)
    # update lag features for next
    days if i < len(test) - 1:
        # update lag1 of the next day to be this prediction
        test.loc[i+1, "lag1"] = p
        # update rolling metrics (ma7, ma14, std7) for next day using recent
        known/predicted values
        recent_meals = final_preds[-6:] + train.loc[max(0, i-
        5):, TARGET].tolist() # last 6
        predicted plus last ... (simplified)
        test.loc[i+1, "ma7"] = np.mean(recent_meals[-
        7:]) if len(recent_meals) >= 7 else np.mean(recent_meals)
        test.loc[i+1, "ma14"] = np.mean(recent_meals[-14:]) if
        len(recent_meals) >= 14 else np.mean(recent_meals)
        test.loc[i+1, "std7"] = np.std(recent_meals[-7:], ddof=0) if
        len(recent_meals) >= 7 else 0.0

# ----- Save submission -----
submission = pd.DataFrame({
    "Date": test_raw["Date"].astype(str).str.slice(0, 10), # format date
    as string
    "Nb_personnes_qui_ont_mang ": final_preds
})

```

```
submission.to_csv(OUT, index=False)
print("✓ submission.csv saved – rows:", len(final_preds))
```

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29  
30 31 32 33 35 36 37 38 39 40 41 42 43 44 45 48 51 52 53 56 57 report of BNP.pdf  
file:///file-NC9e1YgNRNKSggAeiwZp8z  
34 46 47 49 50 54 55 U23\_BNP\_Paribas\_EI\_Djazair.ipynb  
file:///file-67FrtuM6RPCE8WFPmzQZNV